

5 Complex Systems Quiz ANS

1. What does the reductionist paradigm assume?

- A) A system can be understood by understanding its component parts
- B) Components can never be studied
- C) Only social systems exist
- D) Statistics is always invalid

ANS: A) A system can be understood by understanding its component parts

2. What is emergence?

- A) A feature located entirely in one component
- B) A system-wide property not present in any individual component
- C) A linear input-output relation
- D) A manually specified instruction

ANS: B) A system-wide property not present in any individual component

3. What does non-linearity mean?

- A) Outputs always equal inputs
- B) Every effect is proportional
- C) A change in input need not produce a proportional change in output
- D) Feedback is impossible

ANS: C) A change in input need not produce a proportional change in output

4. What is positive feedback?

- A) A process that restores a variable to equilibrium
- B) A process with no circular influence
- C) A one-time external shock
- D) A process that amplifies an initial change

ANS: D) A process that amplifies an initial change

5. What is self-organization?

- A) Components coordinate into collective patterns without centralized explicit instructions
- B) A designer specifies every local action
- C) The system has one component
- D) The system cannot adapt

ANS: A) Components coordinate into collective patterns without centralized explicit instructions

6. What is criticality?

- A) Permanent equilibrium far from change
- B) A system poised near a tipping point between states
- C) Simple additive behavior
- D) The elimination of feedback

ANS: B) A system poised near a tipping point between states

7. What is distributed functionality?

- A) Each concept resides in one clearly labeled component
- B) No component contributes to function
- C) Functions are spread across many components rather than neatly localized
- D) Only the central controller matters

ANS: C) Functions are spread across many components rather than neatly localized

8. What does scalable structure imply?

- A) Doubling size always doubles every property
- B) Power laws never appear
- C) Large systems equal two isolated half-size systems
- D) System properties may scale nonlinearly with size

ANS: D) System properties may scale nonlinearly with size

9. What is adaptive behavior?

- A) Changing behavior in response to environmental demands
- B) Repeating one action regardless of context
- C) Removing all feedback
- D) Preventing learning

ANS: A) Changing behavior in response to environmental demands

10. Why is armchair analysis limited for AI safety?

- A) Theory is useless in all domains
- B) Not all emergent properties and failure modes can be anticipated without experimentation
- C) Experiments reveal nothing
- D) AI systems are completely predictable

ANS: B) Not all emergent properties and failure modes can be anticipated without experimentation

11. What can happen when a complex system develops subgoals?

- A) Subgoals always preserve the original goal
- B) Subgoals disappear immediately
- C) Subgoals can supersede or distort the original objective
- D) The system becomes linear

ANS: C) Subgoals can supersede or distort the original objective

12. Why can scaling an AI system create new risks?

- A) Scaling only reduces capability
- B) Scaling leaves behavior identical
- C) Small systems contain every future property
- D) New emergent capabilities and failure modes may appear

ANS: D) New emergent capabilities and failure modes may appear

13. What lesson follows from the rarity of directly built complex systems?

- A) Scale gradually from simpler, tested systems rather than build a huge system at once
- B) Skip testing until full scale
- C) Novel properties are impossible
- D) Safety is guaranteed by size

ANS: A) Scale gradually from simpler, tested systems rather than build a huge system at once

14. Why is human oversight insufficient by itself?

- A) Humans never make decisions
- B) Human judgment is fallible and processes may be too fast or complex to understand
- C) Humans always detect every hazard
- D) Oversight removes the need for system design

ANS: B) Human judgment is fallible and processes may be too fast or complex to understand

15. What characterizes a wicked problem?

- A) Exactly one correct solution
- B) A fixed system with an obvious stopping condition
- C) Multiple causes, social complexity, intervention risk, and no flawless solution
- D) A solution derivable from given information only

ANS: C) Multiple causes, social complexity, intervention risk, and no flawless solution

16. Why can interventions in complex systems backfire?

- A) Systems have no interactions
- B) Interventions affect one isolated variable only
- C) Every stable state is desirable
- D) Feedback and interdependence can generate unintended consequences

ANS: D) Feedback and interdependence can generate unintended consequences

17. What does a stable state tend to do?

- A)** Resist small attempts to move the system away from it
- B)** Guarantee catastrophe
- C)** Eliminate restoring forces
- D)** Change randomly after every input

ANS: A) Resist small attempts to move the system away from it

18. Which was cited as a partially successful complex intervention?

- A)** Ignoring ozone depletion
- B)** Smallpox eradication
- C)** Eliminating all public health coordination
- D)** One-day climate stabilization

ANS: B) Smallpox eradication

19. What should investigators examine instead of only a single root cause?

- A)** One operator's final action
- B)** One failed component only
- C)** Broader systemic and organizational factors
- D)** Only the system's average output

ANS: C) Broader systemic and organizational factors

20. What overall approach does AI safety require as a wicked problem?

- A)** A single permanent technical fix
- B)** No experimentation
- C)** Ignoring the deployment environment
- D)** Careful sociotechnical management with continuous learning and adaptation

ANS: D) Careful sociotechnical management with continuous learning and adaptation